Status	Finished
Started	Tuesday, 20 October 2026, 4:06 AM
Completed	Tuesday, 20 October 2026, 4:15 AM
Duration	8 mins 54 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search divides the list into halves.
c. Linear search is more efficient than binary search on large lists.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues in the left or right sublist ✓
b. The search continues from the beginning
c. The list is sorted
d. The search stops

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By comparing each element sequentially
c. By dividing the list length by two ✓
d. By using a hash function

Question **4**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. 1
c. n/2
d. n ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O🥐
c. O(n log n)
d. O(1) ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. An error is raised
c. The first element is returned
d. The search is considered unsuccessful ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. At the beginning of the list
c. In the right sublist
d. In the entire list

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By using a hash function
- c. By sorting the list first
- d. By dividing the list into halves

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Dynamic search
- b. Binary search
- c. Linear search ✓
- d. Hashing

Question **10**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list contains duplicate elements
- b. When the list is sorted
- c. When the element is found in the list ✓
- d. When the search algorithm finishes

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Must use a sorted array
- b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- d. There must be a mechanism to access middle element directly

Question **12**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Deleting elements from a list
b. Sorting elements in a list
c. Determining whether an element is present in a list ✓
d. Inserting elements into a list

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted
b. There must be a mechanism to delete and/or insert elements in the list ✓
c. There should be direct access to the middle element in any sublist
d. Number values should only be present

Number values should only be present

Question **14**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Hash search
c. Linear search
d. Binary search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. log n
c. n/2
d. n ✓